

8 Introduction to hypothesis testing

Subject content	Additional information
8.1 Use and demonstrate understanding of the terms parameter, statistic, unbiased and standard error.	Students will be expected to know that: • a parameter is a numerical property of a population • a statistic is a numerical property of a sample and is a function only of the values in the sample and contains no unknown parameters. Students should be familiar with the notations μ and σ^2 for a population mean and variance respectively.
8.2 Know and use the language of statistical hypothesis testing: null hypothesis, alternative hypothesis, significance level, test statistic, 1- tail test, 2-tail test, critical value, critical region, and acceptance region and p -value.	Students will be expected to use the convention of a 5% significance level unless directed otherwise.
8.3 Know that a sample is being used to make an inference about the population and appreciate the need for a random sample and of the necessary conditions.	Students should be aware that a conclusion to a hypothesis test should not be stated as definite.
8.4 Choose the appropriate hypothesis test to carry out in particular circumstances.	
8.5 Conduct a statistical hypothesis test for the proportion in the binomial distribution and interpret the results in context using exact probabilities or, where appropriate, a normal approximation.	In a hypothesis test on a population proportion, candidates may use either π or p as the parameter in their hypotheses.
8.6 Conduct a statistical hypothesis test for the mean of a normal distribution with known or assumed variance, from a large sample, and interpret the results in context.	
8.7 Know the importance of appropriate sampling when using hypothesis tests and be able to critique the conclusions drawn from rejecting or failing to reject a null hypothesis by considering the test performed.	